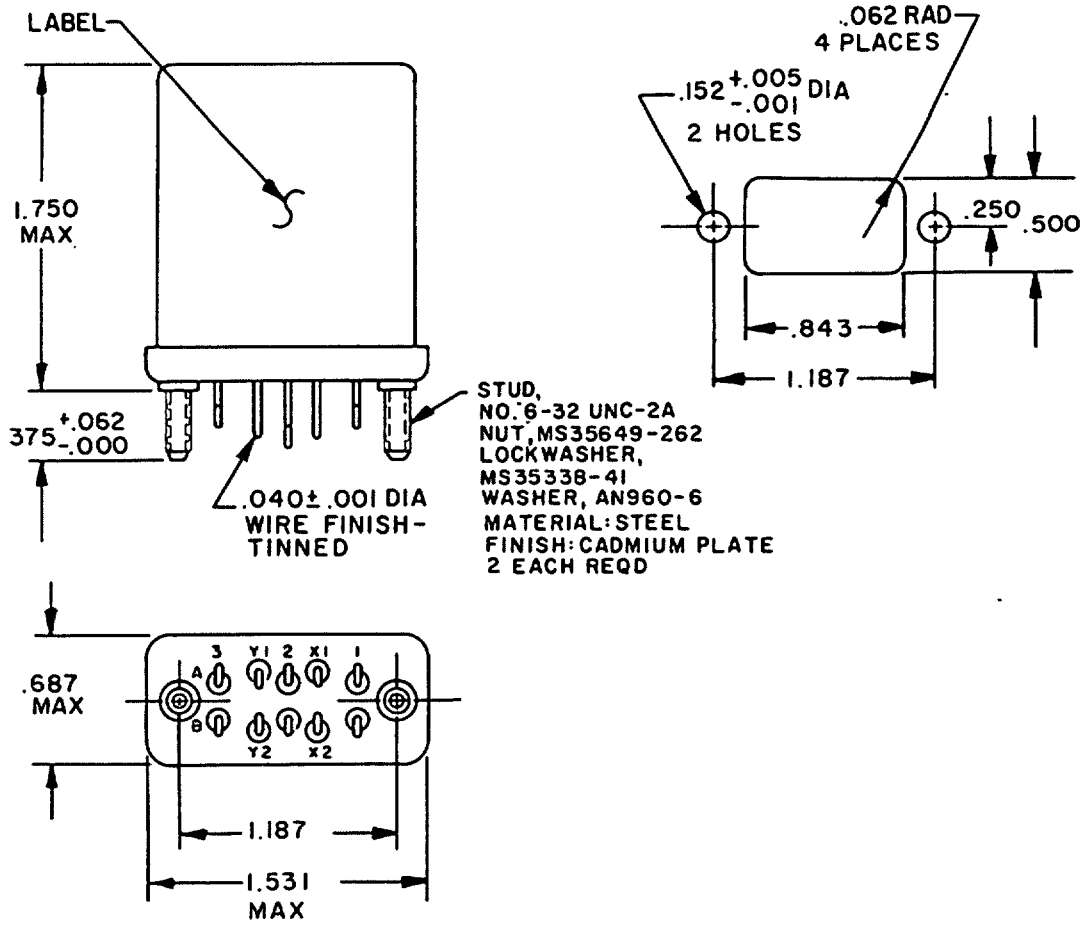
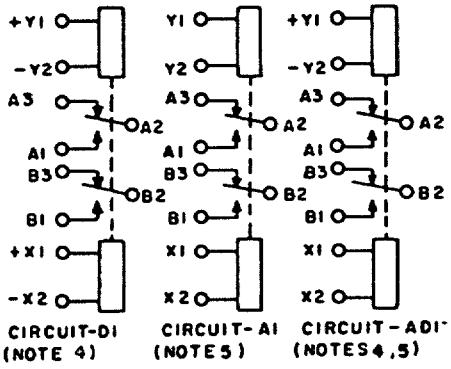
INACTIVE FOR NEW DESIGN AFTER 5 JUN 87
NO SUPERSEDING STANDARD.
(FOR NEW DESIGN USE MIL-R-6106/38)

FED. SUP CLASS
5945



Inches	mm
.000	0.00
.001	0.03
.005	0.13
.040	1.02
.062	1.57
.152	3.86
.172	4.37
.250	6.35
.375	9.53
.500	12.70
.687	17.45
.843	21.41
1.187	30.15
1.531	39.89
1.750	44.45



(D) ENTIRE STANDARD REVISED

APPROVED 1 Mar 1963 REVISED (D) 5 JUN 87

User activities: Army - Navy - Air Force -

Review activities: Army - Navy - EC Air Force - 11

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P.A AF - 85 Other Cust Navy - AS	International Interest	TITLE RELAYS, ELECTROMAGNETIC, 2 PDT, 5 AMPERES, TYPE I, MAGNETIC LATCH, SOLDER TERMINALS, STUD MOUNTED, HERMETICALLY SEALED	MILITARY STANDARD MS25465
Procurement Specification MIL-R-6106		SUPERSEDES:	PAGE 1 OF 5

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AMSC N/A

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NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.010 (0.25 mm).
4. Terminal numbers need not appear on relay headers provided there is affixed to the relay a suitable legible circuit diagram that permanently and positively identifies each terminal location specified herein.
5. The use of diodes on ac relays is optional. Actual application must be shown on label.
6. Relay is magnetically latched in both positions. Caution note to observe polarity must appear on relays with dc coils.
7. Shock, vibration, and acceleration requirements application with coils de-energized.
8. In the event of confilict between the text of this standard and the references cited herein, the text of this standard shall take precedence.
9. Referenced Government documents of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation form a part of this standard to the extent specified herein.

TABLE I. Dash numbers and characteristics.

Dash number MS 25465-	Type	Coil	Terminal type	Mounting or mating socket	Max weight in pounds
D1	I	dc	Solder hook	Stud	0.18
A1	I	ac	Solder hook	Stud	0.19
AD1	I	ac-dc	Solder hook	Stud	0.19

User activities: Army - Navy - Air Force -
Review activities: Army - EC Navy - Air Force - 11
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Procurement Specification MIL-R-6106

P.A AIR FORCE Other Cust NAVY - AS

International Interest

SUPERSEDES:

TITLE RELAYS, ELECTROMAGNETIC, 2 PDT, 5 AMPERES, TYPE 1, MAGNETIC LATCH, SOLDER TERMINALS, STUD MOUNTED, HERMETICALLY SEALED

MILITARY STANDARD

MS25465

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FED. SUP CLASS 5945

TABLE II. Operating characteristics.

MS part no. MS25465-	Coil data												Time - (milliseconds maximum)				
	Coil	Nominal			Max		Max pick-up voltage			Drop-out voltage	Operate 4/	Release 5/	Contact bounce				
		Volts 1/	Freq. Hz	Res Ω	Volts	Amperes	Normal 3/	High temp test	Cont current test				Main		Aux		
													NO	NC	NO	NC	
D1	X1, X2 Y1, Y2	28	dc	N/A	29	0.12	18	18	19.8	N/A	25	N/A	2	2			
A1	X1, X2 Y1, Y2	115	400 2/	N/A	122	0.06	90	90	95	N/A	25	N/A	2	2			
AD1	X1, X2	115	400 2/	N/A	122	0.06	90	90	95	N/A	25	N/A	2	2			
	Y1, Y2	28	dc	N/A	29	0.12	18	18	19.8	N/A	25	N/A	2	2			

1/ CAUTION: Use of any coil voltage less than nominal coil voltage will compromise the operation of the relay.
2/ MS25465-A1 and -AD1 may be used on 60 Hz if maximum ambient temperature is limited to +85°C (maximum coil current shall be 0.066 ampere).
3/ Over the temperature range.
4/ With nominal coil voltage.
5/ From nominal coil voltage.

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P.A AIR FORCE - 85 Other Cust / NAVY - AS

International interest

SUPERSEDES:

TITLE RELAYS, ELECTROMAGNETIC, 2 PDT, 5 AMPERES, TYPE I, MAGNETIC LATCH, SOLDER TERMINALS, STUD MOUNTED, HERMETICALLY SEALED

MILITARY STANDARD

MS25465

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FED. SUP CLASS 5945

TABLE III. Rated contact load (amperes per pole) (case grounded).

Type of load	Life operating cycles X 10^3	28 V dc				115 V ac, 1 phase				115/200 V ac, 3 phase 1/				See appropriate notes
		Main		Aux		Main		Aux		Main		Aux		
		NO	NC	NO	NC	400 Hz	60 Hz	400 Hz	60 Hz	400 Hz	60 Hz	400 Hz	60 Hz	
Resistive	100	5	5			5	4							
Inductive	100													
Inductive	20	3	3			3	2							
Motor	100	1.5	1.5			1.5	1							
Lamp	100	0.8	0.8			0.8	0.6							
Transfer load														2/
Mechanical life reduced current	400	1.25	1.25			1.25	1							
Intmd current	Applicable per specification													

1/ Absence of value indicates relay is not rated for 3-phase applications.
2/ Transfer load indicates relay suitable for transfer between unsynchronized ac power supplies at rating indicated.

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P.A AIR FORCE-85
Other Cust NAVY-AS

International Interest

SUPERSEDES:

TITLE
RELAYS, ELECTROMAGNETIC, 2 PDT, 5 AMPERES, TYPE I, MAGNETIC LATCH, SOLDER TERMINALS, STUD MOUNTED, HERMETICALLY SEALED

MILITARY STANDARD

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Environmental characteristics

Temperature range	-70°C to +125°C
Max altitude rating	80,000 ft
Shock G-level	50 G
Duration	11 ms
Max duration contact opening	10 µs
Vibration - sinusoidal	
G-level	10 G
Frequency range	5 - 1500 Hz
Acceleration	15 G

Electrical characteristics

Insulation resistance, initial	100 megohms
After life or environmental tests	50 megohms

Dielectric strength initial (sea level)

	Initial	After life tests
Coil to case	1,000 V rms	1,000 V rms
Aux contacts		
All other points	1,000 V rms	1,000 V rms

Dielectric strength (altitude)	(When mounted in mating socket) 80,000 ft
Coil to case	250 V rms
Aux contacts	
All other points	250 V rms

Max contact drop initial	0.150 volt
After life test	0.175 volt
Overload current	20 amperes
Rupture current	25 amperes
Duty rating	Continuous
RFI specification	MIL-STD-461

(Applicable to coil circuits of ac operated relays)

Quality conformance inspection

Performance of groups B and C testing are not applicable.

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